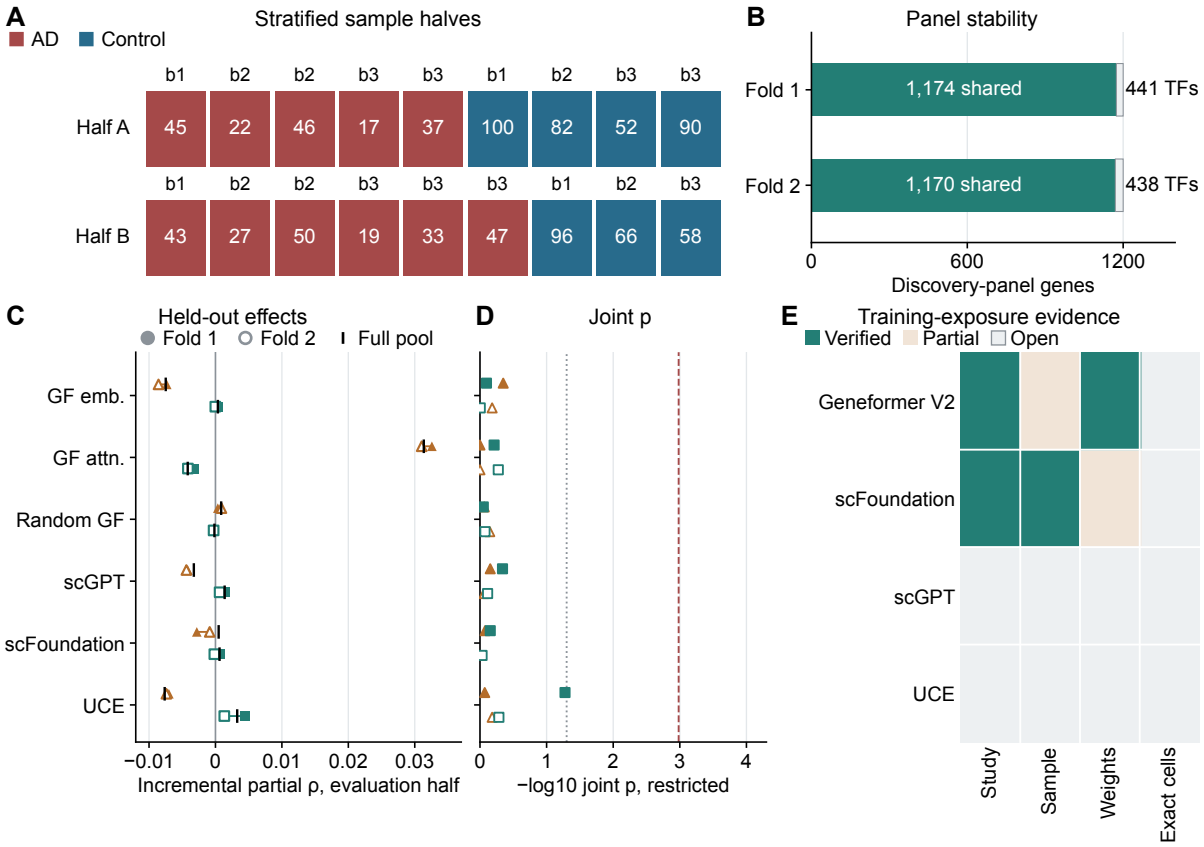
A
Stratified sample halves
AD
Control
b1 b2 b2 b3 b3 b1 b2 b3 b3
Half A
45 22 46 17 37 100 82 52 90
b1 b2 b2 b3 b3 b3 b1 b2 b3
Half B
43 27 50 19 33 47 96 66 58
B
Panel stability
Fold 1
1,174 shared
441 TFs
Fold 2
1,170 shared
438 TFs
0
600
1200
Discovery-panel genes
C
Held-out effects
Fold 1
Fold 2
Full pool
GF emb.
GF attn.
Random GF
scGPT
scFoundation
UCE
−0.01
0
0.01
0.02
0.03
Incremental partial ρ, evaluation half
D
Joint p
0
1
2
3
4
−log10 joint p, restricted
E
Training-exposure evidence
Verified
Partial
Open
Geneformer V2
scFoundation
scGPT
UCE
Study
Sample
Weights
Exact cells